

Task 6: Introduction to Cryptography

Tools:

- Primary: OpenSSL
- Alternatives: CyberChef

Hints / Mini Guide:

1. Learn symmetric vs asymmetric encryption.
2. Encrypt files using AES.
3. Generate RSA keys.
4. Understand digital signatures.
5. Hash files and verify integrity.
6. Compare encryption algorithms.
7. Learn real-world usage (HTTPS, VPN).
8. Document findings.

Deliverables:

- Cryptography experiment report

Final Outcome:

- Strong cryptography fundamentals

Interview Questions Related To Above Task:

- What is encryption?
- Symmetric vs asymmetric encryption?
- What is a hash?
- What is digital signature?
- Why is cryptography important?

📌 Task Submission Guidelines

- 🕒 **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🔧 **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💰 **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

- 📌 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [[Submission Link](#)]

Best
of
Luck

